

Sub-Sec	Subject	Qs	Marks
d-1	Civil Engg. Courses	5	5
d-2	Textile Engg. Courses	5	5
d-3	Chemical/Printing Technology	2	2
d-4	Ceramic Engineering	2	2
d-5	Food Technology	2	2
d-6	Others (Agri/Architecture/Fashion)	4	4
TOTAL		20	20

d-1 CIVIL ENGINEERING COURSES (5 Questions | 5 Marks)

■ Building Materials

- Bricks: raw materials (clay, sand, lime, iron oxide); manufacturing: moulding→drying→burning (Clamp/Hoffman kiln)
- Brick work: English bond, Flemish bond; foundation brick work; efflorescence causes
- Cement types: OPC, PPC, PSC, rapid hardening, low heat, sulphate resistant; grades 33/43/53
- Concrete: grades M15/M20/M25/M30; workability (slump test); water-cement ratio; compaction methods
- Rocks classification: igneous (granite, basalt), sedimentary (limestone, sandstone), metamorphic (marble, slate)

■ Soil Mechanics

- Index properties: void ratio e , porosity n , degree of saturation S , water content w
- Seepage: Darcy's law $q=kiA$; permeability k ; flow net

■ Water Supply

- Per capita demand: 135 lpcd (domestic); sources: surface water (rivers, lakes), groundwater (wells, springs)
- Treatment: sedimentation → coagulation → filtration → chlorination
- Sludge disposal: drying beds, anaerobic digestion, land fill

■ Foundation & Structures

- Foundation types: shallow (isolated, combined, strip, raft/mat) vs deep (pile, caisson, well)
- RCC: Fe250, Fe415, Fe500 steel; M15-M40 concrete; cover requirements
- Repair & maintenance: crack filling, repointing, waterproofing, corrosion treatment

■ Surveying

- Basics: plan vs map; scales; chain surveying (Gunter's chain=20m, Engineer's chain=30m)
- Bench Mark: permanent (GTS, arbitrary) and temporary; levelling staff, dumpy level
- Compass surveying: Prismatic compass (WCB 0-360°), Surveyor's compass (QB N/S 0-90° E/W)

■ Roads, Bridges & Irrigation

- Roads: flexible pavement (bituminous) vs rigid (CC); layers: subgrade, sub-base, base, surface
- Bridges: classification — major/minor; RCC/PSC/steel/suspension; site selection criteria
- Irrigation: duty, delta, base period; methods: surface (flood, furrow), sprinkler, drip

- Dams: gravity, arch, buttress, earthen; reservoir site: geological stability, impermeability
- Earth moving machinery: bulldozer, scraper, grader, excavator, roller, dump truck

■ Doors, Roofs & Ecology

- Door types: flush, panelled, glazed, louvred, sliding, folding; material: timber, steel, UPVC
- Roofs: flat, pitched (lean-to, couple, purlin, truss), curved; suitability by rainfall/span
- Ecology: ecosystem, biodiversity, water pollution (sewage, industrial effluents); non-conventional energy: solar, wind, biogas, geothermal

d-2 TEXTILE ENGINEERING COURSES (5 Questions | 5 Marks)

- Fibres: Natural (cotton, silk, wool, jute, linen) vs Synthetic (nylon, polyester, acrylic, rayon)
- Yarn manufacturing: opening & cleaning → carding → drawing → roving → spinning (ring spinning, OE rotor)
- Fabric: woven (plain, twill, satin weaves), knitted (weft/warp), non-woven (bonded)
- Weaving Technology: loom types — power loom, shuttle-less (rapier, air-jet, water-jet, projectile)
- Textile Testing: tensile strength (lea strength, count strength product), elongation, crimp, twist, evenness (CV%)
- Quality Control: statistical tools, control charts, grading, pilling, abrasion resistance testing
- Modern Yarn Production: compact spinning, air-jet spinning, friction spinning; core-spun yarns
- Bleaching: removes natural color; agents — H_2O_2 , NaOCl , sodium hydrosulphite (for wool)
- Dyeing: fibre/yarn/fabric stage; vat dyes (cotton), acid dyes (wool), disperse dyes (polyester)
- Printing: block, screen (flat/rotary), digital inkjet, transfer printing; discharge/resist styles

d-3 CHEMICAL / PRINTING TECHNOLOGY (2 Questions | 2 Marks)

- Fluid Flow: laminar vs turbulent; $\text{Re} = \rho v D / \mu$; Bernoulli equation; Newtonian fluids; pumps & compressors
- Chemical Process Industry: fertilizers, petroleum refining, cement, paper, soap & detergents, pharmaceuticals
- Agro-based Industries: sugar (crushing, clarification, evaporation, crystallization), distilleries, edible oil extraction
- Petrochemicals: LPG, naphtha, benzene, ethylene, propylene; cracking (thermal/catalytic); reforming
- Printing Machines: platen press, cylinder press, web offset press; major types — letterpress, offset lithography
- Types of Printing: relief (letterpress), planographic (offset litho), intaglio (gravure), screen (serigraphy), digital

d-4 CERAMIC ENGINEERING (2 Questions | 2 Marks)

- Classification: traditional ceramics (pottery, bricks, tiles, glass, cement) vs advanced/technical ceramics (Al_2O_3 , SiC , ZrO_2)
- Pottery: earthenware (porous, low temp), stoneware (dense, medium temp), porcelain (vitrified, high temp, white)
- Ceramic Materials: silica SiO_2 , alumina Al_2O_3 , feldspar, kaolin (clay); properties — hardness, brittleness, heat/corrosion resistance
- Properties: high compressive strength but low tensile; excellent thermal and electrical insulation; chemical inertness; refractoriness

d-5 FOOD TECHNOLOGY (2 Questions | 2 Marks)

- Vitamins: fat-soluble (A, D, E, K) vs water-soluble (B-complex, C); deficiency diseases — A (night blindness), D (rickets), C (scurvy), B1 (beriberi), B3 (pellagra)

- Cereals: wheat (gluten; chapati/bread), rice (starch; staple), maize, barley, sorghum; storage pests; milling processes
- Pulses: high protein; lentils, chickpeas, moong, urad; dal milling; antinutritional factors (lectins, tannins)
- Milk & Milk Products: pasteurization (72°C/15sec HTST; 63°C/30min LTLT); homogenization; cream, butter, cheese, paneer, ghee, condensed/dried milk
- Food Preservation: physical (heat — canning/pasteurization; cold — refrigeration/freezing), chemical (salt, sugar, vinegar, preservatives BHA/BHT), biological (fermentation, pickling), radiation (irradiation)

d-6 OTHER ENGINEERING COURSES (4 Questions | 4 Marks)

■ Agriculture

- Farm equipment: tractor (2WD/4WD), plough (MB/disc/chisel), harrow, cultivator, seed drill, thresher, combine harvester
- Power sources: draught animals, tractors, electric motors; PTO; hydraulic systems on tractor

■ Architecture

- History of Indian Architecture: Indus Valley (Mohenjo-daro — grid planning, sanitation); Mauryan (Ashoka pillars, Sanchi Stupa); Gupta (temples); Mughal (Taj Mahal, Red Fort, Fatehpur Sikri); Colonial (Indo-Saracenic, Gothic Revival)
- Building Typologies: residential, commercial, industrial, institutional, recreational, religious, mixed-use; site planning; zoning

■ Fashion Design & Technology

- Fashion Technology: fabric selection, pattern making, grading, marker making, cutting, sewing, finishing
- History & Culture: haute couture, prêt-à-porter, fashion cycles; Indian traditional textiles — Banarasi, Kanjeevaram, Bandhani, Phulkari (Haryana)
- Garment Manufacturing: sewing machines — lock stitch (SNLS), chain stitch, overlock; presser feet, needles; industrial vs domestic machines

FULL EXAM SUMMARY

Section	Topic	Questions	Marks
A	Basic Sciences	25	25
B	Electronics Stream	25	25
C	Mechanical Stream	20	20
D	Other Streams	20	20
GRAND TOTAL		90	90

Duration: 90 Minutes | No Negative Marking | 1 Mark per Question | MCQ (4 options, 1 correct)